CHEMISTRY - STANDARD 9

Chapter: 2

Periodic Table

1) How many elements are included in the Modern Periodic table which is widely used at present?
(a) 110 b) 112 c) 118 d) 63

2) How many periods are present in the modern periodic table?
a) 10 b) 7 c) 18 d) 14

3) In how many groups are elements distributed in the modern periodic table?
a) 7 b) 14 c) 10 d) 18

4) Which period has the least number of elements?
a)7 b) 2 c) 6 d) 1

5) How many electrons are present in the outermost shell of group - 1 elements?
a) 7 b) 1 c) 2 d) 3

6) Which is the group number of Rubedium (37 Rb)?
a) 3 b) 7 c) 8 d) 1

7) Halogens are included in--------
a) group18 b) group 2 c)group 7 d) group 17

8) The group number of alkali metals is----------

9) The number of electrons present in the outermost shell of group II elements are--------

10) Elements of group 18 are known as ----------

11) Choose the correct statement that is applicable to Group-1 elements.
a) Show similarity in chemical properties
b) The number of outermost electrons are different.
c) There is one electron in the outermost shell
d) There are two electrons in the outermost shell

12) Assertion and Reason type
Assertion (A) : Elements of group II show similarity in chemical properties.
Reason(R) : They have same number of electrons in their outermost shell.
a) Assertion and Reason are true, but R is not the correct explanation of A
b) Assertion and Reason are true,R is the correct explanation of A
c) A is true but R is false
d) A is false, but R is true

13) What is the main reason why main group elements in periods 2 and 3 have similar properties?
- A. They have the same atomic numbers
- B. They have the same electron configurations
- C. They have the same physical properties
- D. They have the same atomic masses

14) Which group in the periodic table includes alkali metals like Sodium (Na) and Potassium (K)?
- A. Group 1
- B. Group 2
- C. Group 17
- D. Group 18

15) Which element is an example of a metalloid?
- A. Sulfur (S)
- B. Silicon (Si)
- C. Sodium (Na)
- D. Chlorine (Cl)

16) Elements are arranged in Mendeleev's periodic table on the basic of ----------

17) Select the modern periodic law from the given below.
i. The chemical and physical properties of elements are periodic functions of their atomic mass.
ii. The chemical and physical properties of element are periodic functions of number of neutrons.
iii. The chemical and physical properties of elements are periodic functions of their atomic numbers

18) The scientist who prove properties of elements depend mainly on atomic number is....
(Rutherford, J J Thomson, Henry Moseley)

19) Incomplete electron configuration of an element belongs to 4th period and 13th group is given below. Complete it and find its atomic number.
Electron configuration 2, 8, 18, ...
(18, 3 ,1)

20) To which group do Alkali metals belong?
(1, 2, 3)

21) Second group elements are commonly known as -------
(Alkali metals, Alkaline earth metals, Noble gases)

22) To which group group do halogens belong?
(15, 16, 17)

23) To which group group do noble gases belong?
(16, 17, 18)

24) Transition elements belong .....group ?
(1 and 2 groups, 3rd to 12th group, 13th to 18th group)

25) Which of the following elements is a solid at room temperature?
(Oxygen, Hydrogen, Sodium)

26) Maximum number of electron that can accumulate in the outer most shell is....
(8, 18, 32)

27) How many elements are there in third period in the modern periodic table?
(8, 18, 32)

28) Electron configuration of Helium element belongs to 4th period and 1st group.
(2, 4; 2, 8, 1; 2, 8, 8, 1)

29) Which of the following elements do not belongs to main group?
(1 and 2 groups, 3rd to 12th group, 13th to 18th group)

30) Write the electron configuration and identify family name

Elements	Electron configuration	Group	Family Name
$_{11}Na$			
$_{20}Ca$			
$_{17}Cl$			
$_{10}Ne$			

31) In transition elements electron filling takes place in -------- shell
(Outer most shell, penultimate shell, in any incomplete shell)

32) How many elements are there in the 6th and 7th period
(8, 18, 32)

33) The distance between the centre of nucleus of an atom to the outermost shell in which electron occupies is known as.....
(Atomic diameter, Atomic radius, Atomic volume)